

Sitraka Randrianarivo

Clemson, SC — srandri@clemson.edu

EDUCATIONAL BACKGROUND

- Expected **Ph.D. in Mathematical Sciences**
May 2027 Clemson University. Advisor: Dr. Felice Manganiello
- 2023 **M.S. in Mathematical Sciences**
Clemson University
- 2021 **M.Sc. in Mathematical Sciences**
African Institute for Mathematical Sciences (AIMS) South Africa
- 2018 **Bachelor's Degree in Pure Mathematics**
Faculty of Sciences, University of Antananarivo, Madagascar

PROFESSIONAL APPOINTMENTS

Clemson University, School of Mathematical and Statistical Sciences

- 2025–present **Instructor of Record**
Math 1010: Essential Math for an Informed Society (Fall 2025)
Math 1040: Precalculus and Introduction to Differential Calculus (Spring 2026)
Math 1060: Calculus I (Fall 2026).
- 2022–present **Graduate Teaching Assistant**
Linear Algebra (Math 3110); College Algebra (Math 1980);
Calculus I (Math 1060); Calculus of Several Variables (Math 2060);
Business Calculus I (Math 1020); Business Calculus II (Math 2070)

PUBLICATIONS

- [1] Felice Manganiello and Sitraka Randrianarivo. Subspace code construction over the complex numbers using extended multiplicative characters. *Submitted*, 2026.
- [2] Felice Manganiello and Sitraka Randrianarivo. Exact cardinality and nonredundant parametrization of character-polynomial codes. *Submitted. arXiv:2607.11595*, 2026.
- [3] Sitraka Randrianarivo. Rational Reconstruction for Homological Construction. Master's thesis, African Institute for Mathematical Sciences, South Africa, 2021.

HONORS AND AWARDS

- 2021 **Full Scholarship for M.S. Study**
African Institute for Mathematical Sciences (AIMS) South Africa.

RESEARCH TALKS AND POSTERS

- [1] Subspace codes from finite fields for the operator channel over complex number. In *2026 Joint Mathematics Meetings (JMM 2026)*. AMS, 2026.
- [2] Subspace codes for Analogue Operator Channel, 2025. Special-session talk, 2025 Spring Southeastern Sectional Meeting, Clemson University, Clemson, SC.
- [3] Construction of optimal codes for sum-rank weight, 2024. Poster presentation, Data sECurity and mAchine Learning (DECAL) Workshop, Clemson University, Clemson, SC.

RESEARCH MENTORSHIP

Graduate Research Mentor / TA, Clemson University

Summer 2022 Research Experience for Visiting Undergraduate (Math 4830).
Co-guided a group of visiting undergraduate researchers
through original investigations in sum-rank codes.

LANGUAGES

English: Fluent **French:** Fluent **Malagasy:** Native

OUTREACH AND COMMUNITY ENGAGEMENT

2026 **Volunteer**, Clemson Calculus Challenge

2024, 2026 **Math-In Volunteer**, Clemson University
Helped students prepare for final exams by answering questions on Calculus of
Several Variables (Spring 2024), Precalculus, and Calculus I (Spring 2026).

TECHNICAL SKILLS

Mathematical typesetting	L ^A T _E X
Computer algebra systems	SageMath, Macaulay2, Singular, Mathlab
Programming	Python, C